TITLE PAGE

- **Problem Statement ID –**
- **Problem Statement Title- Autonomous Production Choke Controller for a Single Naturally Flowing Oil Well**
- **Theme-**
- **PS Category- Software**

IDEA TITLE

Proposed Solution (Describe your Idea/Solution/Prototype)

- Detailed explanation of the proposed solution
- How it addresses the problem
- Innovation and uniqueness of the solution

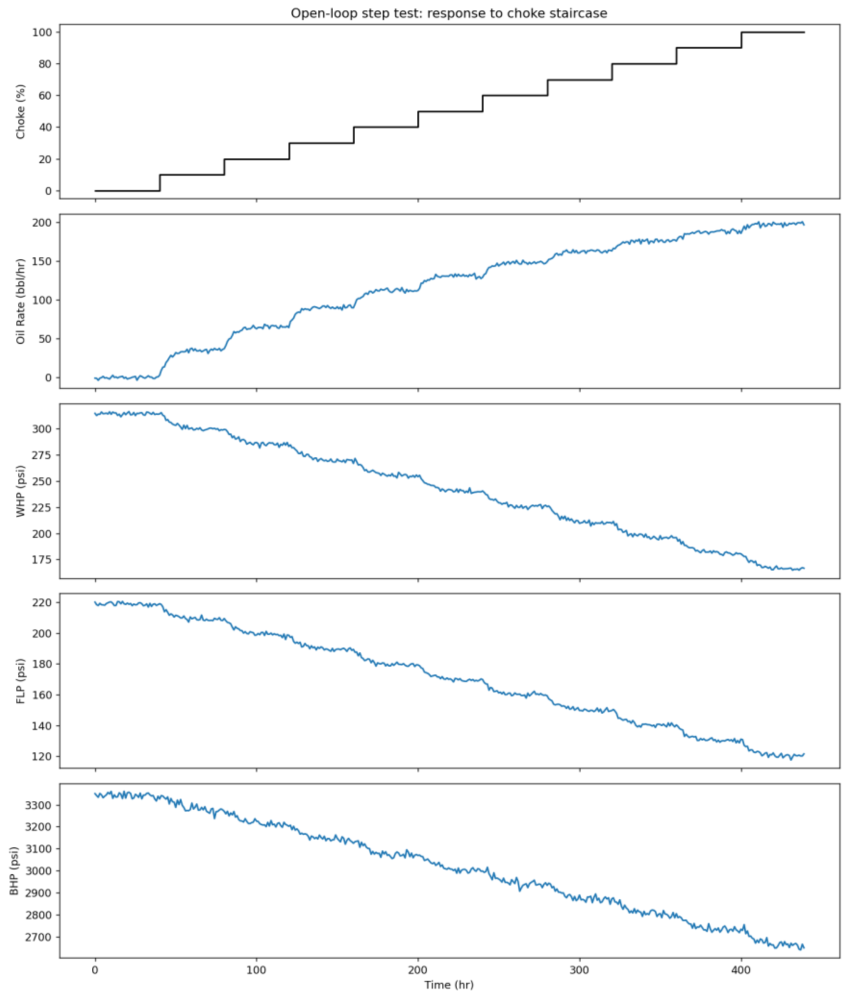
TECHNICAL APPROACH

- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

TECHNICAL APPROACH — PROCESS UNDERSTANDING & MODEL (I)

- No official simulator was provided — data/simulator.py is a calibrated substitute fit to the reference CSV (30–65% choke tested).
- Steady-state map y_ss(u) = A + B·u/(u+uh) per channel, driving first-order-plus-dead-time dynamics with exact zero-order-hold discretization.
- A fresh open-loop step test — our own staircase experiment, not the reference CSV — identifies the model the controller predicts with.
- A timing bug in the fitting code was found and fixed: dead time now matches ground truth exactly in all 20 seed/channel checks (right).

Channel	True τ	Ident. τ	Error	Range
Q	5.00	4.95	3.0%	0.0–5.0%
WHP	7.50	7.20	4.0%	0.0–6.7%
FLP	7.00	6.95	3.6%	0.0–7.1%
BHP	9.00	8.25	8.3%	2.8–13.9%



TECHNICAL APPROACH — PROCESS UNDERSTANDING & MODEL (II)

- Hybrid physics + learned-correction layer, kept per channel only where it demonstrably improves held-out accuracy — not applied by default.
- WHP/FLP/BHP safety limits are one-sided (floor-only or ceiling-only), matching the brief's own description of each variable's risk direction: WHP ≥ 205 psi, BHP ≥ 2830 psi, FLP ≤ 200 psi.
- Two monitored-but-not-constrained channels — Wellhead Temperature and Annulus Pressure — add situational awareness without ever entering the safety filter.

Channel	Physics-only RMSE	+Correction RMSE	Kept?
Q	1.66	1.63	✅ used
WHP	1.30	1.30	❌ skipped
FLP	0.92	0.89	✅ used
BHP	9.85	9.81	✅ used

TECHNICAL APPROACH — CONTROL STRATEGY (I): THE ALGORITHM

- A one-step receding-horizon search evaluates 11 candidate choke moves (±5% per interval) every control cycle — named precisely, not oversold as a full trajectory-optimizing MPC.
- Each candidate is held constant over a derived lookahead horizon and simulated forward with the identified model.
- Feasible candidates are ranked by closeness to target production, with a move-suppression term that prevents valve chatter once near target.

TECHNICAL APPROACH — CONTROL STRATEGY (II): SAFETY GUARANTEES

- Every candidate is screened against the safety envelope, tightened by 3σ of identified measurement noise, before being considered.
- If nothing fully satisfies every limit, the controller falls back to the move minimizing predicted violation — it never freezes.
- Empirically strong, not formally guaranteed: there's no recursive feasibility proof that today's safe choice keeps one available tomorrow.
- Example — Scenario C (400 bbl/hr, unreachable): settles at 162.8 bbl/hr @ 66% choke, the max the envelope allows, 0/100 violations.

TECHNICAL APPROACH — CONTROL STRATEGY (III): EXPLAINABILITY

- Every choke decision logs a one-sentence, human-readable rationale — a real output field, not just a design principle — directly answering operator distrust of opaque automation.
- Dashboard/output structure echoes Honeywell Uniformance's PHD / KPI / Asset Sentinel layering: a small-scale proof-of-concept of the same wellhead-to-control-room problem Honeywell's UPPS addresses at scale (echoes, not replicates).

TECHNICAL APPROACH — ACTUATOR ACTIVITY & CHATTERING FIX

- Move-suppression fix: candidates were originally ranked by predicted target error alone, so once near target, noise-sized differences between near-tied candidates chattered the valve.
- Added cost = target_error + λ·|move size| (λ = 1.0 bbl/hr required per %-point moved) to both the normal and safety-fallback branches:

Scenario	Moves	Total valve travel
A	5 / 80	16.0 %-pts
B	7 / 140	29.0 %-pts
C	53 / 100	129.0 %-pts (was 154.0 pre-fix)

FEASIBILITY AND VIABILITY

- Analysis of the feasibility of the idea
- Potential challenges and risks
- Strategies for overcoming these challenges

ARTIFACTS

- Relevant artifacts, such as :
  - Copy of the Code Embedded
  - Snaps of the solution proposal
  - Dashboard snaps

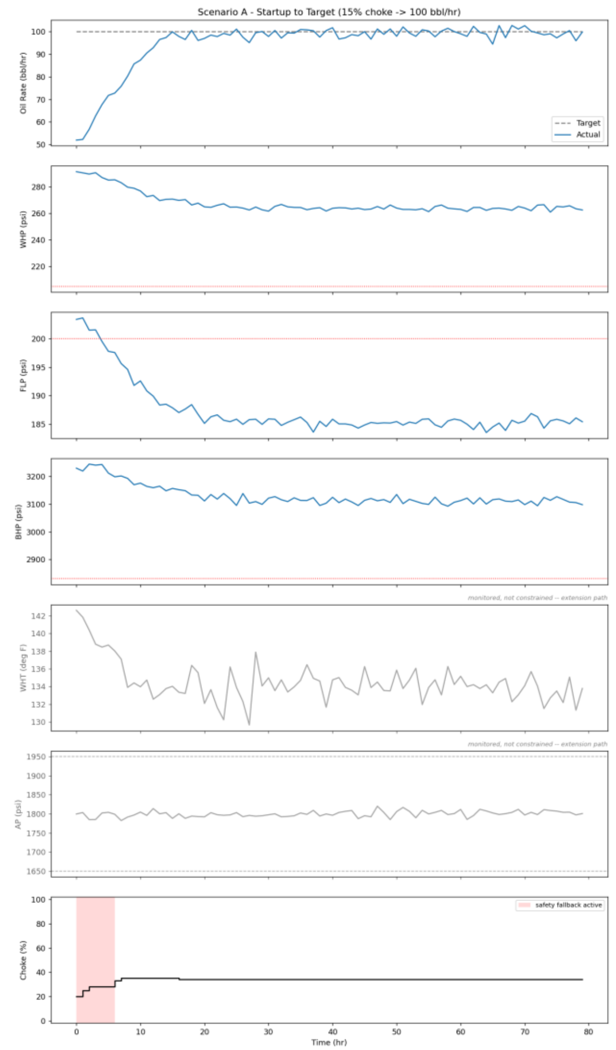
ARTIFACTS — RESULTS: SCENARIO OVERVIEW

- Scenario A — Startup to Target: 15% choke → 100 bbl/hr target.
- Scenario B — Target Tracking: ~34% choke start, 100 → 150 bbl/hr step.
- Scenario C — Infeasible Target: ~34% choke start, 400 bbl/hr requested (deliberately unreachable).
- Full time-series trends for A/B/C, and Scenario D, follow on their own slides.

Scenario	Setup	Final	Constraint violations
A — Startup to Target	15% choke, 80h run	99.9 bbl/hr @ 34% choke	4/80
B — Target Tracking	34.2% choke, 140h run	150.4 bbl/hr @ 61% choke	0/140
C — Infeasible Target	34.2% choke, 100h run	160.8 bbl/hr @ 66% choke	0/100

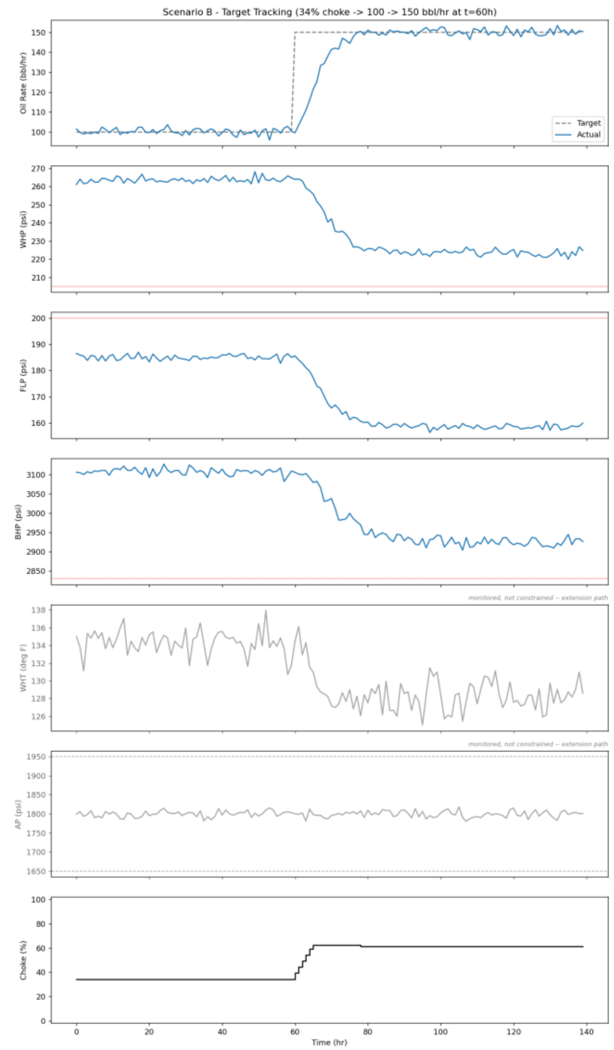
ARTIFACTS — RESULTS: SCENARIO A — STARTUP TO TARGET

- 15% choke start, 100 bbl/hr target, 80h run. Trailing 10h mean settles at 99.3 bbl/hr @ 34% choke.
- FLP starts above its own steady-state ceiling at 15% choke, so early violations are a deterministic consequence of the ramp-rate limit, not sensor noise or model guesswork.
- Safe by hour 4; 30-seed sweep: violations in every seed (mean 3.97/80), always resolved within 3–5 hours.



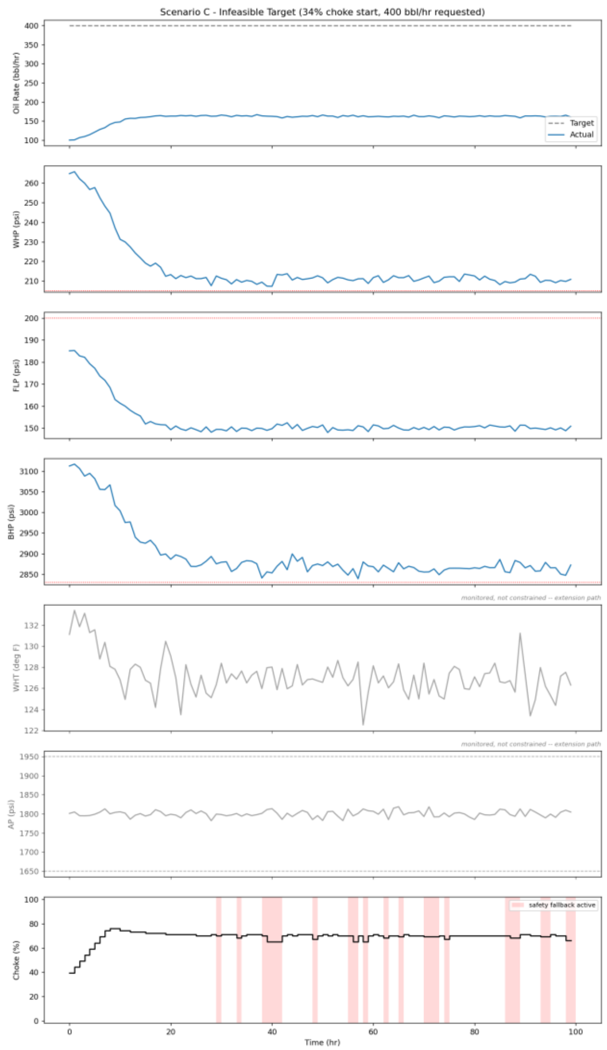
ARTIFACTS — RESULTS: SCENARIO B — TARGET TRACKING

- 34% choke start, target steps 100 → 150 bbl/hr at hour 60, 140h run. Settles at 150.6 bbl/hr @ 61% choke.
- Zero constraint violations in the representative run and in all 30 seeds — the only scenario with a perfectly clean safety sweep.
- Clean because it never leaves the calibration's supported 30–65% choke band.



ARTIFACTS — RESULTS: SCENARIO C — INFEASIBLE TARGET

- 34% choke start, 400 bbl/hr requested — deliberately unreachable.
- The controller does not chase it into a violation: settles at 162.8 bbl/hr @ 66% choke, the maximum the tightened safety envelope allows.
- 0/100 violations in the representative run and in all 30 seeds.



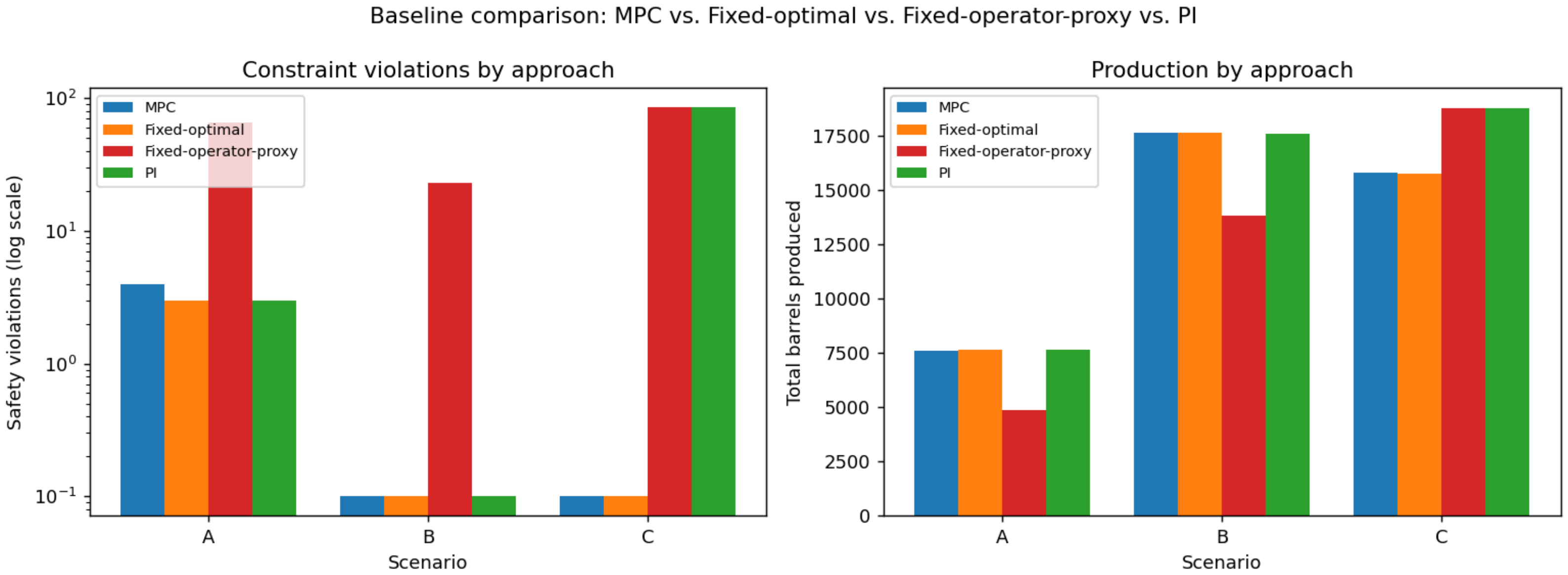
ARTIFACTS — RESULTS: SEED-SWEEP SAFETY DISTRIBUTION

- 30-seed distribution per scenario — not a single cherry-picked run.
- Reports mean/max constraint violations and safety-fallback frequency per scenario.

Scenario	Mean violations	Max violations	Seeds w/ ≥1 violation	Mean fallback rate
A — Startup to Target	3.97 / 80	5	30 / 30	8.58%
B — Target Tracking	0.00 / 140	0	0 / 30	0.00%
C — Infeasible Target	0.00 / 100	0	0 / 30	23.57%

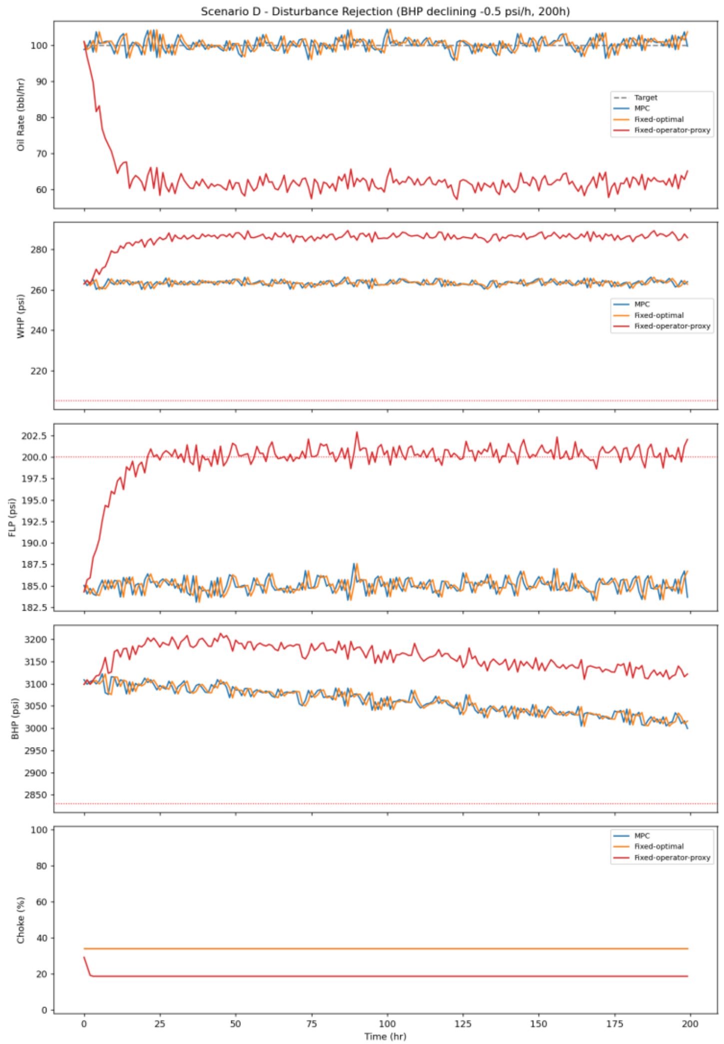
ARTIFACTS — RESULTS: BASELINE COMPARISON

- MPC vs. Fixed-optimal vs. Fixed-operator-proxy vs. PI — identical scenarios, model, limits, and simulator seeds.
- Fixed-operator-proxy models a real operator with no model and no envelope knowledge: a naive linear read off the raw reference CSV, backed off 15 points.
- Headline: MPC and Fixed-optimal both stay safe in every scenario; the operator-proxy and safety-blind PI both rack up major violations chasing Scenario C's infeasible target.



ARTIFACTS — RESULTS: SCENARIO D — DISTURBANCE REJECTION

- Deliberate relaxation of “no changing reservoir properties”: BHP's steady-state offset drifts −0.5 psi/h for 200h at a constant 100 bbl/hr target.
- Tests whether live re-planning beats a static setpoint once the plant genuinely drifts.
- Result: MPC ties Fixed-optimal — BHP and Q are uncoupled here, so re-planning has nothing extra to correct. The real, consistent win is envelope-awareness itself vs. the operator-proxy (right).



Approach	Violations	Barrels	1st violation
MPC	0/200	20,040.9	never
Fixed-optimal	0/200	20,048.3	never
Fixed-op.-proxy	121/200	12,594.2	21h

RESEARCH AND REFERENCES

- Details / Links of the reference and research work